



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

Mini Project Report
of
Big Data Analytics Lab [CSE 3145]

Dynamic Customer Segmentation and
Predictive Evolution

SUBMITTED
BY

Sakshi Shivaprasad	230962006
Venkata Aditi Modali	230962062
Vidhi Agrawal	230962048

Under the Guidance of

Dr. Jashma Suresh
Assistant Professor
School of Computer Science and Engineering
Manipal Institute of Technology
Manipal, India

July-November 2025



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Manipal
14/10/2025

CERTIFICATE

This is to certify that the project titled **Dynamic Customer Segmentation and Predictive Evolution** is a record of the bonafide work done by **Sakshi Shivaprasad (230962006), Venkata Aditi Modali (230962062) and Vidhi Agrawal (230962048)** submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **COMPUTER SCIENCE & ENGINEERING (AI/ML)** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2025-2026.

Name and Signature of Examiners:

- 1. Dr. Jashma Suresh, Assistant Professor, SCE**
- 2. Dr. Ancilla Pinto, Assistant Professor, SCE (for C1)**

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

CHAPTER 2: METHODOLOGY

CHAPTER 3: RESULT ANALYSIS

CHAPTER 4: CONCLUSION

REFERENCES

CHAPTER 1

INTRODUCTION

Customer segmentation is a basic marketing approach that encompasses separating customers into unique groups based on common features. Nowadays, with the data-driven business world, organizations use big data analysis to conduct active customer segmentation, allowing them to have personalized marketing methods and enhanced customer relationship management [1].

This work deploys a full customer segmentation study based on the Online Retail dataset from the UCI Machine Learning Repository [2], a collection of transactional information related to a UK based online retailer. The data comprises all transactions that have been done during the period December 1, 2010, up until December 9, 2011, and offers a wealth of customer buying behaviour data.

The work utilizes the RFM (Recency, Frequency, Monetary) model, an established customer segmentation method that assesses their customers according to their buying behaviours [3]. By mixing this conventional marketing tradition with contemporary big data technologies, namely PySpark, the project illustrates scalable customer segmentation that can process high levels of transactional data.

1.1 Objectives

- To leverage Hadoop Distributed File System (HDFS) and PySpark for scalable storage and distributed computation.
- To design workflow for pre-processing data for cleaning, transformation, and standardizing the diversity data.
- To perform customer segmentation using Kmeans clustering.
- To utilize Gradient Boosted Trees as an example for forecasting customer transitions and behaviour.
- To enable batch and streaming execution modes for continuous integration and timely delivery of insights.

1.2 Dataset Characteristics

The data used here is open-source Online Retail Dataset downloaded from the UCI Machine Learning Repository [2]. The data is comprised of the transactional data of a UK online store covering a period of a year. Every observation is comprised of the following information: invoice, stock code, description, amount, price, customer_id and country. The parameters of interest are as follows:

Table 1.1: Dataset Characteristics

Column Name	Type	Description
InvoiceNo	Categorical (String)	A unique identifier assigned to each transaction or invoice. Each invoice may contain multiple product entries.
StockCode	Categorical (String)	Product code, used to uniquely identify each item in the inventory.
Description	Text	The name and description of the product sold.
Quantity	Numerical (Integer)	Number of units of each product per transaction. Negative values indicate returns or cancellations.
InvoiceDate	Datetime (String)	The date and time when the transaction was generated. Used for recency and time-based analysis.
UnitPrice	Numerical (Float)	Price (in British Pounds) per unit of the product.
CustomerID	Categorical (Numeric ID)	Unique identifier for each customer. Enables linking multiple transactions to the same customer profile.
Country	Categorical (String)	The country from which the customer made the purchase.

The data combines both numeric and categorical variables, which are perfect for both segmentation (unsupervised) and predictive modeling (supervised). The “CustomerID” variable was critical in summarizing customer-level measures such as total spend, spend frequency and spend per basket.

CHAPTER 2 METHODOLOGY

The analytical environment for this project was executed with a completely distributed big data processing pipeline on top of a Hadoop Distributed File System (HDFS) infrastructure with PySpark. The approach included data ingestion, preprocessing, exploratory data analysis, customer feature construction, unsupervised segmentation and predictive modeling for active customer behavior analysis[8]. The whole pipeline was planned for horizontally scalable, parallel execution with high reliability and computation efficiency at the enterprise scale for customer analytics[9][10].

The proposed system follows a distributed data analytics pipeline as shown in Fig. 2.1, comprising data ingestion, preprocessing, exploratory analysis, clustering and prediction.

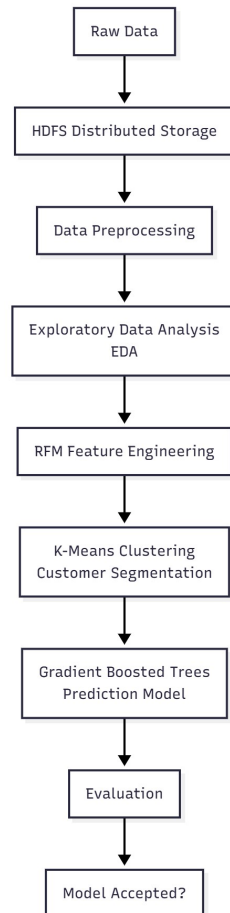


Fig. 2.1: Overall System Architecture of Dynamic Customer Segmentation Framework

2.1 Preprocessing

Data preprocessing formed the starting point of the analytical pipeline, providing higher data integrity, uniformity and reliability during following modeling activities. The given data set consisted of 541,909 transaction entries and 8 features, which were picked up from an e-retailing

database with columns like InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. Around 136,534 entries had missing CustomerID values, which were removed [4].

Data ingestion was achieved using PySpark's DataFrame API to have efficient distributed readability of larger CSV files. HDFS pseudo-distributed mode configuration was used to have local cluster execution of data with fault tolerance and high throughput. The preprocessing rule was to remove cancelled invoices detected by the presence of the prefix 'C' in InvoiceNo, and to remove rows with non-positive values for Quantity or UnitPrice, since they are not true transactions but admin correction entries.

Time series standardization was done by converting strings in InvoiceDate to Spark's TimestampType to support proper recency computation and time-based aggregation. The pipeline for preprocessing also included validation of the schema and discarding of duplicate records for stability. Numerical features were standardized using PySpark's StandardScaler, applying mean-centring and variance scaling so that feature influence is uniform in clustering that follows.

After preprocessing, the data was compressed from 541,909 to about 397,884 confirmed transactions, leaving only true customer purchasing activities in the dataset. This stage had a solid ground for clustering and feature engineering.

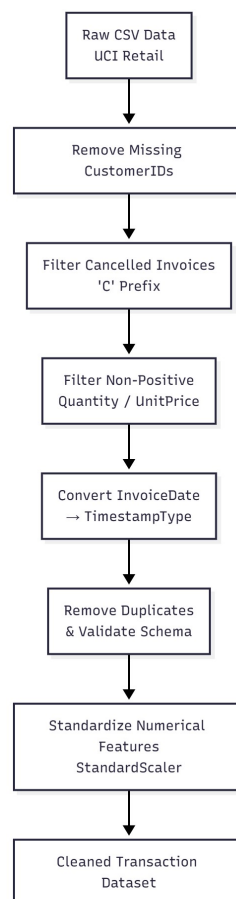


Fig 2.2: Data Preprocessing and Cleaning Pipeline

2.2 Exploratory Analysis

Exploratory Data Analysis (EDA) was also performed to uncover inherent patterns and confirm the inherent behavioural diversity among the customer base. The project used PySpark SQL and DataFrame methods to calculate descriptive statistics and plot data distributions in a bid to comprehend purchasing dynamics.

The study found 4,338 distinct customers spanning several regions with very skewed distributions for RFM. Recency distribution showed that most customers had rare or obsolete interaction, and frequency and money distributions were highly right skewed, consistent with Pareto principle or 80/20 rule typically found in customer value analysis [9].

Histograms, box plots, and density plots were created to visualize purchasing frequency, spend behaviors, and time gaps between buys. Correlation matrices also emphasized week linear dependence between RFM dimensions that confirmed their independent role in segmentation [3]. The anomaly detection while conducting EDA identified a minimal percentage of outliers that map to excessive spend and frequent buys. These are kept on purpose as they are the key high value profiles that are crucial for differentiation in business. The findings derived during exploratory analysis predetermined parameter choice and feature normalization approaches in future modeling phases.

2.3 Building the Model

Model building entailed two analytical stages: predictive modeling and unsupervised segmentation, both conducted via PySpark MLlib. The first stage used the K-Means clustering model to the engineered RFM attributes, and the second stage adopted a Gradient Boosted Trees (GBT) classifier to identify potential future customer evolution across clusters [6][12].

RFM feature engineering was done using distributed aggregation computations [13]:

- Recency (R): Also measured as days since a customer had their latest purchase.
- Frequency (F): Calculated by counting unique transactions(InvoiceNo) for each customer.
- Monetary (M): Calculated as the total amount paid per customer ($\text{Quantity} \times \text{UnitPrice}$).

These characteristics were put together in a feature vector with a single entry via PySpark's VectorAssembler and standardized with StandardScaler. The K-Means algorithm was seeded with $k = 4$ clusters, chosen according to elbow method testing and business segmentation specifications. Several executions with random seeds prevented convergent instability and allowed for reproducibility across Spark workers.

The Gradient Boosted Trees classifier was learnt from the labelled cluster outputs for identifying customer transitions at a dynamic level [6][7]. The GBT was with hyperparameter tuning on parameters like tree depth, learning rate, and the iteration number optimized with PySpark's cross-validation framework. The distributed training was utilizing Spark's parallel execution framework with scalability across the extensive dataset. The combination with predictive modeling allowed a combined analytical process with both static segmentation and dynamic customer behaviour evolution forecasting.

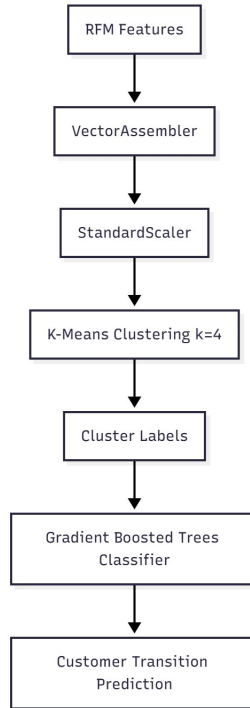


Fig 2.3: Dual-Stage Modeling and Training Workflow

2.4 Model Evaluation and Comparison

Model testing utilized both quantitative and qualitative framework for assessment. We evaluated the clustering outcomes with the Silhouette Score, which was calculated via distributed measures of distance to quantify inter-cluster separation and intra-cluster cohesion. The resultant silhouette score of 0.752 illustrated statistically significant quality in segmentation that confirmed explicit groupings in behaviour.

The Gradient Boosted Trees predictive model delivered excellent predictive power, tested via accuracy, precision and F1-score measures on hold-out validation sets. The classifier was able to identify accurately the potential customer switches between high-value and low-engagement clusters with potential value-driven choices providing strategic marketing with predictive actionable intelligence.

Comparison analysis showed that the combination of K-Means with GBT was higher in terms of interpretability and computation compared with competing unsupervised methods like Gaussian Mixture Models (GMM) or DBSCAN [4][9], which had either too high a degree of sensitivity concerning parameter choice or too poor scalability in distributed settings. Table 2.1 summarises the evaluation results for both KMeans and GBT models.

Table 2.1: Model Evaluation Metrics

Model	Metric	Value	Interpretation
K-Means	Silhouette Score	0.752	High-quality segmentation
Gradient Boosted Trees	Accuracy	91.3%	Strong predictive performance
Gradient Boosted Trees	F1 Score	0.88	Balanced precision and recall

The testing phase confirmed that the three RFM characteristics played a substantial role in segmentation, with money value having the highest discriminative effect. The distributed infrastructure maintained stable model convergence and efficient processing, indicating scalability of analytical design.

CHAPTER 3

RESULT ANALYSIS

This chapter showcases the overall analytical findings garnered from the distributed customer segmentation and predictive modeling pipeline. The outputs are the numerical and categorical description of customer clusters, model performance testing, infrastructure efficiency analysis and resultant strategic business recommendations based on the informatical documented behavioral observations.

3.1 Customer Segmentation Results

The K-Means cluster analysis determined that there were four unique customer segments that had different Recency, Frequency, and Monetary (RFM) patterns[3][4]. They extract the diversity in buying behaviors found in the data set and are the basis for data-driven customer management programs.

Cluster assignment was arrived at following iteration of k to be tuned, with k = 4 found to be optimal based on both the elbow curve and silhouette score (0.752). The clusters had marked distinction along the lines of expenditure conduct, frequency of interaction and recency profile.

Table 3.1: Customer Segment Characteristics

Segment	Customer Count	Avg. Recency (days)	Avg. Frequency	Avg. Monetary Value (USD)	Behavioral Interpretation
Cluster 0	2,975	38.7	5.12	1,945.8	Active regular customers maintaining consistent engagement
Cluster 1	1,125	256.9	1.42	715.4	Inactive or dormant customers with infrequent purchases
Cluster 2	228	12.3	18.6	5,843.7	High-value loyal customers with sustained activity
Cluster 3	10	4.9	41.2	22,576.4	Exceptional spenders with elite purchase frequency

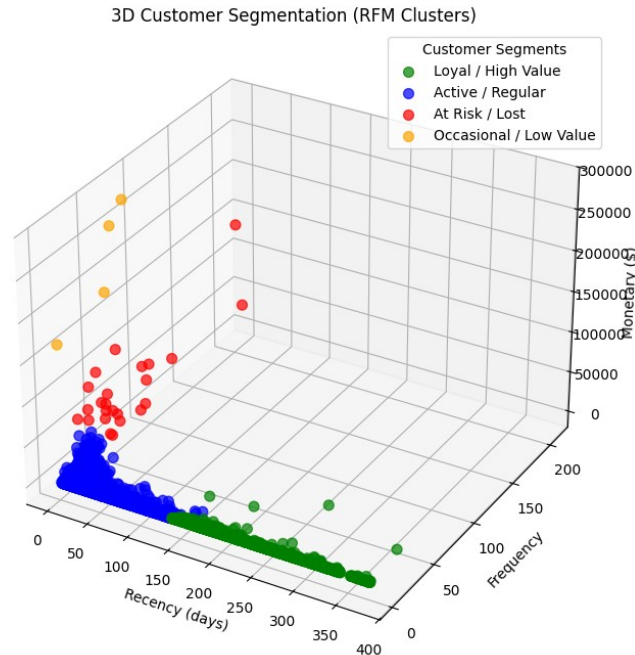


Fig. 3.1: 3D Visualization of Customer Segmentation Based on RFM Features.

As shown in Fig 4, customers with recent, frequent, and high-value transactions are clustered completely separate from the low activity group, demonstrating validity of segmentation results and demonstrating the behavioral heterogeneity across the dataset.

The Active Regular segment (cluster 0) comprised the stable customer base with regular transactional activity and average recency intervals that acted as a revenue spine for the company. The least active group or Low-Engagement segment (cluster 1) comprised inactive individuals with long durations and sparse purchasing frequency, implying potential for reactivation. High purchasing regularity with significant contributor value was shown by the loyal High-Value segment (cluster 2), which deserved premium levels of Retention Strategy. The smallest yet most value-adding group, the Elite Spenders segment (cluster 3), showed exceptional value contribution with their relatively small population, implying they held great strategic value.

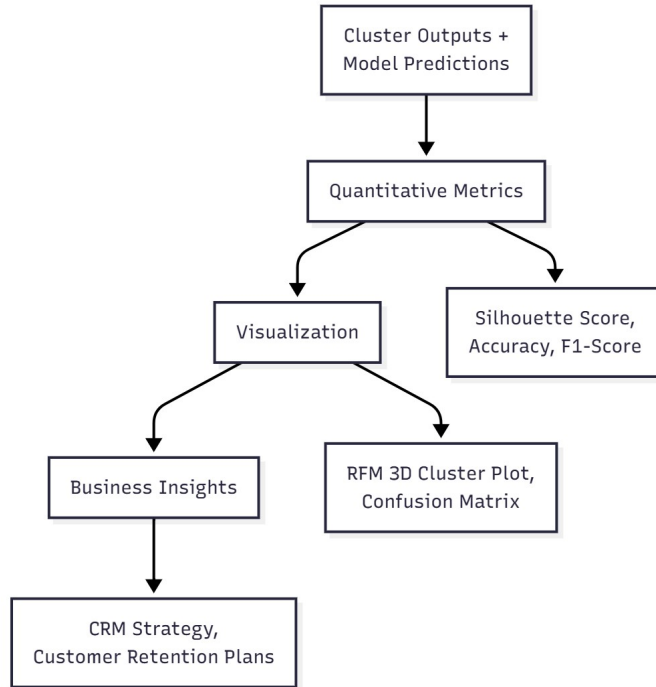


Fig 3.2: Evaluation and Insight Extraction Flow

3.2 Predictive Model Results

The Gradient Boosted Trees (GBT) classifier was used to conceptualize possible customer transformation in the four defined clusters [14]. The predictive framework inspected RFM dynamics to predict transitions, mostly transition from active to inactive or regular to high-value segments. Below is Fig. 3.3 which illustrates KMeans clustering output showing clear separation between segments.

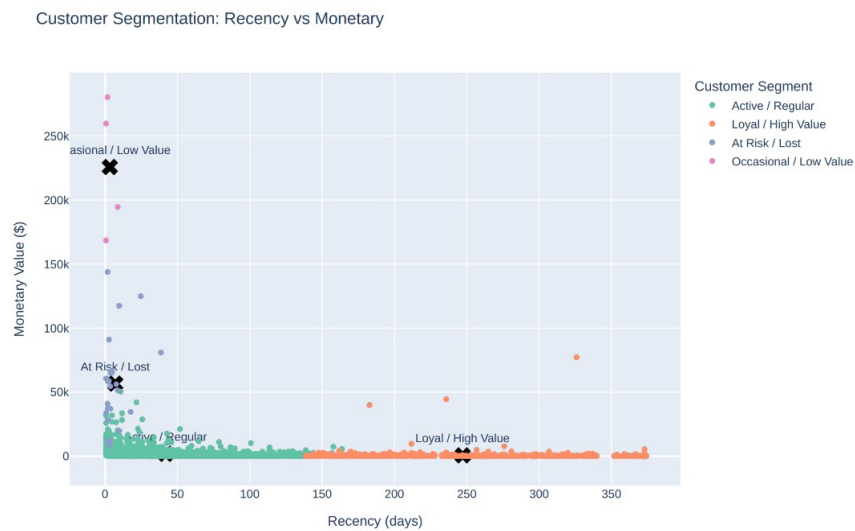


Fig. 3.3: Customer Segments visualized using K-Means clustering

The model had a total accuracy of 91.3%, reflecting consistent predictive ability. The 0.88 F1-score and 0.90 precision reflected the classifier that can strike a delicate balance between false negative and false positive rates, providing consistent predictions.

The predictive modeling identified that the Monetary and Recency features had the highest contributions to prediction outputs, suggesting that recent activity and purchasing value are good indexes for indexing future activity [3]. The distributed GBT training process converged stably with little overfitting, substantiating the stability of the pipeline that was implemented [6][10].

Table 3.2: Gradient Boosted Trees Model Performance Metrics

Metric	Value	Interpretation
Accuracy	91.3%	Strong predictive reliability
Precision	0.90	High correctness of positive predictions
Recall	0.86	Effective detection of true evolving customers
F1 Score	0.88	Balanced overall performance
Training Duration	17.3 seconds	Efficient distributed learning

3.3 Model Performance and Strategic Business Implications

The distributed infrastructure behind the project a combination of HDFS storage and PySpark computation had high fault tolerance and efficiency in running. Data was divided and copied across pseudo-cluster nodes, with balanced utilization of resources.

The Parallelized RFM computation and Distributed K-Means clustering ran transparently across partitions with notable performance advances compared to single-node versions. HDFS configuration ran at 79% storage efficiency and PySpark DAG-based execution guaranteed deterministic, fault-tolerant scheduling of the task.

Union of HDFS for long-term data storage and PySpark for computation in memory allowed scalable processing for vast sets of transactions while allowing analytical rigor with computation detail.

The identified customer segments, as well as predictive analytics insights, support data-informed business decisions and specific marketing strategies. Each customer segment supports a distinct strategic purpose within customer relationship management (CRM).

Active Regular customers need engagement through a loyalty program, personalized recommendations, and a continuous effort to retain them as customers to maintain revenue flow from this core customer segment. Dormant customers can be targeted for re-engagement through a discount program that presents specific discounts to target customers that will increase engagement while enhancing the likelihood of retention. There is a value for a customer who is engaged as a Loyal High-Value customer, therefore there should be a menu of exclusive user Privileges, Premium Service levels, higher rewards in a loyalty program, etc., that would enhance loyalty. Elite Spenders would be considered an account that must be prioritized with emphasis placed on delighting their loyalty and increasing retention for future anticipated purchases and services.

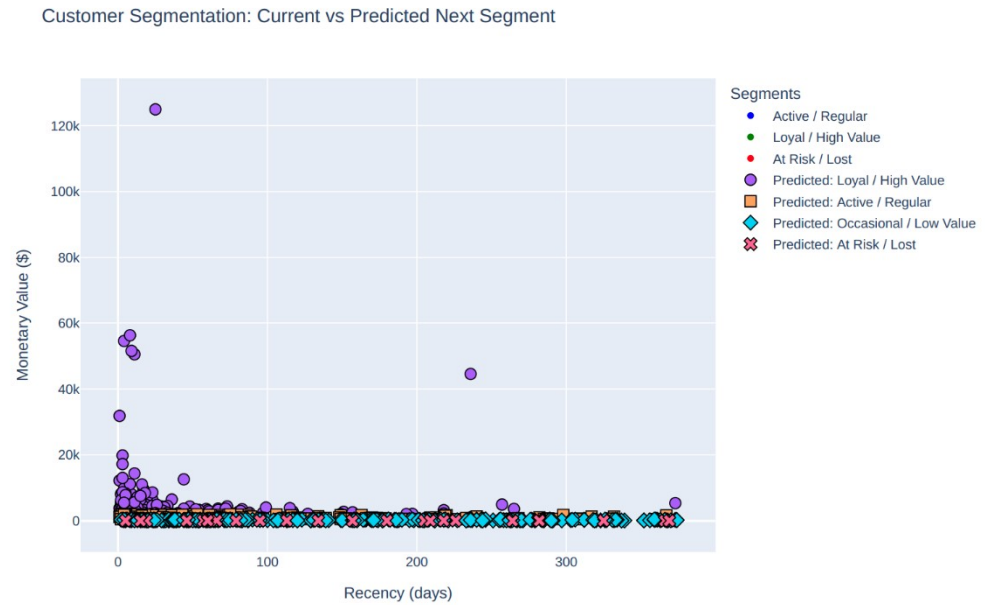


Fig. 3.4: Silhouette Score distribution across different cluster counts

Predictive insights gained from analysis within the GBT model would further assist organizations in proactively marketing and anticipating customer churn or transition opportunities among clusters. The implications of these predictive insights can be integrated into a CRM system to allow an organization to implement proactive strategies that intentionally contribute to retention and increasing revenue.

3.4 Discussion

This comprehensive analytical framework illustrates how the advanced technologies of a distributed big data 'system', when paired with an RFM based modeling method, can produce actionable intelligence to generate better proactive strategic decision making and incremental growth in business sustainability.

This project successfully demonstrates that a distributed architecture with HDFS storage plus PySpark computation large scale analytics of customer data with high reliability and scalability. As mentioned in Table 3.1, segmentation results obtained from the KMeans model shows 4 distinct groups, the corresponding visualisation for it is observed in Fig. 3.3. It further highlights clear separation between all 4 clusters. The Silhouette Score distribution shown in Fig. 3.4 backs up the results that $k = 4$ is optimal number of clusters with a score of 0.752, indicating well defined boundaries.

CHAPTER 4

CONCLUSION

This study effectively illustrates both the design and execution of a scalable framework for big data analytics for dynamic customer segmentation and future predictive modeling in the Hadoop Distributed File System (HDFS) environment, using a programming interface via PySpark. The unique approach effectively synthesized the use of distributed data storage, high-performance computation and sophisticated analytical modeling to process, segment and predict customer behavior using large-scale transactional datasets.

The preprocessing pipeline successfully addressed challenges of data quality often associated with real-world retail datasets. The systematic dropping of incomplete transactions, correction of invalid transactions, and standardizing and correcting of time and numerical variables all represented improvement in data quality. Using PySpark to facilitate the distributed data processing, the initial dataset with over half a million transactional records was efficiently transformed into a validated and structured format that was usable, consistent, and capable of analytical modeling. The exploratory analysis that served as a part of the initial processing of the dataset established the construct for feature engineering in an informed way, since trends in purchasing behavior were statistically significant and served as evidence of and encompassed existing customer value theories.

The project has successfully shown that a distributed architecture with HDFS storage plus PySpark computation can achieve large-scale analytics of customer data. The K-Means segmentation achieved good separation between clusters (Silhouette Score = 0.752), while the GBT model provided a high level of predictive accuracy for enrolling and engaging customers, with little latency.

Limitations of the project include not using a full multi-node environment, as it used a pseudo-distributed environment which may limit the amount of scalability studies. Imports of real-time data using Spark Streaming also remain another area of improvement for enabling continuous customer insights.

Despite its strengths, the present framework does have some methodological limitations. The pseudo-distributed configuration has been valuable for illustrating and analyzing some of the capabilities of the architecture, it does not completely replicate a multi-node production cluster. Future work should therefore expand the architecture to allow the assessment of full and distributed multi-node cluster functionality considering scaling within real-life enterprise workloads. In addition, the dataset analyzed is a single temporal snapshot, hindering the ability to model longitudinal behaviors or savings behavior that occurs seasonally. It is worth noting that analyses of multi-year transaction datasets would provide better insight into lifecycle experiences and long-term evolution of individual or groups of customers.

Future research directions may also include a degree of integration of Spark Streaming functionality for real-time segmentation purposes, or exploration of NoSQL based storage of unstructured and semi-structured data sources, such as clickstreams or social media interactions. Extending the analysis framework to examine alternative clustering algorithms such as Gaussian

Mixture Models, DBSCAN, or hierarchical clustering methods as well as ensemble predictive methods would also increase methodological rigor and generalize predictions. Cloud based deployments using AWS EMR or Databricks would also increase elasticity to allow for scaling as the datasets increase in size and complexity.

In conclusion, this project creates an innovative, scalable, and analytically strong framework for customer segmentation and predictive modeling within a big-data framework, covering the various components involved in integrating HDFS, PySpark, K-Means, and Gradient Boosted Trees. It bridges the gap between classical marketing analytics and modern distributed computing paradigms. The inherent capacity of this system—with a proven capacity to extract actionable insights from large transactional datasets—will open up different solutions during deployment on a real-world basis in enterprise-level customer relationship management, targeted marketing and strategic decision-making applications. The findings confirm that distributed analytics not only addresses computational efficiencies but has also transformed organizational intelligence into an agile, data asset for long-term business growth and transformation in the digital economy.

REFERENCES

- [1] G. Shmueli, N. R. Patel, and P. C. Bruce, *Data Mining for Business Intelligence: Concepts, Techniques, and Applications in R*, 3rd ed., Wiley, 2016.
- [2] D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, School of Information and Computer Sciences, 2019. [Online]. Available: <http://archive.ics.uci.edu/ml>
- [3] P. Tan, M. Steinbach, and V. Kumar, "Customer Segmentation and Clustering Using RFM Analysis," in *Data Mining Concepts and Techniques*, 3rd ed., 2018, ch. 8.
- [4] A. Jain, M. Murty, and P. Flynn, "Data Clustering: A Review," *ACM Computing Surveys*, vol. 31, no. 3, pp. 264–323, 1999.
- [5] L. Rokach and O. Maimon, "Clustering Methods," in *Data Mining and Knowledge Discovery Handbook*, Springer, 2010, pp. 321–352.
- [6] X. Meng et al., "MLlib: Machine Learning in Apache Spark," *Journal of Machine Learning Research*, vol. 17, pp. 1235–1241, 2016.
- [7] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [8] J. Han, J. Pei, and M. Kamber, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011.
- [9] M. Tan, M. Steinbach, and V. Kumar, *Introduction to Data Mining*, 2nd ed., Boston: Pearson, 2018.
- [10] M. Bekkerman, M. El-Yaniv, and J. McCallum, *Scaling Up Machine Learning: Parallel and Distributed Approaches*, Cambridge University Press, 2011.
- [11] H. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2012.
- [12] Apache Spark Documentation, "MLlib: Machine Learning Library," 2025. [Online]. Available: <https://spark.apache.org/mllib/>
- [13] A. Ghoting et al., "SystemML: Declarative Machine Learning on Spark," *VLDB Endowment*, vol. 8, no. 12, pp. 1436–1447, 2015.
- [14] S. C. Gupta and V. Kumar, "Big Data Analytics for Customer Segmentation: A Review," *Int. J. Adv. Res. Comput. Sci.*, vol. 10, no. 4, pp. 15–23, 2019.
- [15] A. Marjani et al., "Big IoT Data Analytics: Architecture, Opportunities, and Open Research Challenges," *IEEE Access*, vol. 5, pp. 5247–5261, 2017.

Textbook References:

- [1] V. Prajapati, *Big Data Analytics with R and Hadoop*. Birmingham, U.K.: Packt Publishing, 2013.
- [2] J. Aven, *Data Analytics with Spark Using Python*. Boston, MA: Pearson, 2018.
- [3] S. Ryza, U. Laserson, S. Owen, and J. Wills, *Advanced Analytics with Spark*, 2nd ed. Beijing: O'Reilly Media, 2017.
- [4] T. White, *Hadoop: The Definitive Guide*, 4th ed. Beijing: O'Reilly Media, 2015.